

Post-Quantum Security of the Sum of Even-Mansour

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Abstract

The Sum of Even-Mansour (SoEM) construction was proposed by Chen et al. at Crypto 2019. This construction implements a pseudorandom permutation via the modular addition of two independent Even-Mansour structures and can spawn multiple variants by altering the number of permutations or keys. It has become the design basis for some symmetric schemes, such as the nonce-based encryption scheme CENCPP* and the nonce-based message authentication code scheme nEHTm. This paper establishes the post-quantum security of the SoEM21 construction under the Q1 model. As a specific variant, SoEM21 employs two independent public permutations alongside a single, uniform cryptographic key. We prove that when an attacker is restricted to classical queries against the keyed construction while maintaining quantum access to the underlying permutations, SoEM21 guarantees up to $n/3$ bits of security. This exactly matches the complexity $O(2^{n/3})$ of the quantum key recovery attack in the Q1 model recently proposed by Li et al., thus establishing a tight bound.

Keywords: Post-Quantum Cryptography, Sum of Even-Mansour, Q1 Quantum Security, Tight Security Bound

1 Introduction

The continuous evolution of quantum computers and quantum algorithmic theory has profoundly impacted the foundational security assumptions of modern cryptography. Over the past decade, as fault-tolerant quantum computing technology rapidly advances from theoretical physics to engineering reality, proving the security of both cryptographic algorithms and symmetric constructions in a quantum computing environment has transitioned into an imperative research trajectory. Early foundational work in this domain was systematically conducted by Boneh et al. [1], who decisively demonstrated that if a quantum adversary is capable of issuing quantum superposition queries to a random oracle, traditional security proofs relying exclusively on the classical Random Oracle Model (ROM) would no longer remain mathematically sound. This profound cryptographic insight catalyzed the formalization of the Quantum Random Oracle Model (QROM), thereby establishing a robust new theoretical framework for post-quantum provable security.

Building upon this paradigm shift, Zhandry [2] subsequently proposed the formal security model for “Quantum-secure Pseudorandom Functions (QPRF),” successfully proving that classical symmetric architectures, such as the GGM tree construction, retain their theoretical security under this augmented model. This pivotal work extended the security guarantees of core cryptographic primitives into quantum querying scenarios, effectively laying the groundwork for synthesizing more complex post-quantum protocols. A major theoretical breakthrough subsequently occurred when Zhandry [3] introduced compressed oracle techniques, enabling the construction of a dynamically updatable quantum state simulation mechanism operating within the Fourier domain. For the first time in cryptographic literature, this elegant mechanism effectively recorded an adversary’s quantum query history without triggering a catastrophic quantum measurement collapse. By overcoming the fundamental “recording barrier” inherent in the QROM, it solidified the basis for critical security reductions, such as proving the quantum indistinguishability of cryptographic hash functions.

Simultaneously, quantum extraction and rewinding techniques—initially developed by Unruh within the context of quantum zero-knowledge proofs [4]—alongside the formulation of the One-way to Hiding (O2H) Lemma [5], provided indispensable analytical tools for precisely quantifying the statistical discrepancies introduced by randomized operations in quantum environments. Furthermore, the measure-and-reprogram framework, meticulously formalized by Don et al. [6], enabled the seamless algorithmic migration of many classical “reprogramming”-based security proofs into rigorous quantum settings. Grilo et al. [7] significantly advanced this methodology by achieving tight adaptive reprogramming bounds within the QROM, supplying more concise and rigorous security proofs for a multitude of key symmetric constructions. Driven by the gradual refinement of these fundamental theories and mathematical tools, three mainstream analytical paradigms for quantum security proofs have currently coalesced within the academic community:

First, the **QPRF-based Substitution Method** relies on systematically replacing the random oracle with an instantiated quantum-secure pseudorandom function, subsequently proving their computational indistinguishability under the adversary’s

defined quantum access capabilities within the reduction sequence. Second, **Compressed Oracle-based Amplitude Analysis** focuses on meticulously quantifying the adversary’s query amplitude directed at specific coordinate points, enabling the reduction to safely resample or rewrite oracle responses in a strictly controlled, unobservable manner. Third, the **Measure-and-Reprogram-based Adaptive Reprogramming Technique** allows for the safe reprogramming of explicit coordinates that the adversary is projected to query with high probability, strictly under the mathematical premise of ensuring a bounded and controllable perturbation probability across the entire query transcript. These sophisticated methodologies have been continuously integrated and optimized, evolving into a generalized, highly adaptable proof framework capable of evaluating complex mechanisms like the Fiat-Shamir transform, advanced digital signatures, Even-Mansour-like block ciphers, and various intricate symmetric cryptographic structures under distinct quantum adversarial threat models.

Pseudorandom functions constructed from public permutations represent a widely ubiquitous design choice in modern cryptographic schemes due to their structural elegance, operational efficiency, and highly transparent security evaluation metrics. Consequently, they stand as prime candidate architectures for standardized primitives in the impending post-quantum era. The Sum of Even-Mansour (SoEM) construction, formally proposed by Chen et al. [8] at Crypto 2019, represents a quintessential and highly influential class of unkeyed pseudorandom functions built directly upon public permutations. It has notably served as the unshakeable foundational building block for advanced symmetric schemes, heavily influencing the architecture of the nonce-based encryption scheme CENCPP* [9] and the nonce-based enhanced message authentication code nEHTm [10]. The classical security limitations of the SoEM21 construction—a specific structural variant utilizing a single uniform key distributed across dual independent public permutations—have been exhaustively established. Existing literature rigorously dictates that its classical security is strictly bounded at the well-known birthday bound, necessitating that an adversary expend a query complexity strictly on the order of $O(2^{n/2})$ to effectively distinguish the construction from a purely ideal random function.

However, with the rapid maturation of quantum cryptanalysis, evaluating the absolute resilience of these fundamental symmetric constructions under diverse quantum models has emerged as the new vanguard of cryptographic research. Within quantum adversarial frameworks, adversary operational capabilities are conventionally bifurcated into two primary threat models: Q1 and Q2 [11]. In the deeply adversarial **Q2 model**, the attacker is theoretically granted the extreme capability to issue quantum superposition queries simultaneously to both the underlying public permutations and the dynamically keyed construction itself. This implies the adversary possesses the operational latitude to execute custom quantum unitaries against the key-dependent oracle, such as seamlessly applying the operator $U_k : |x\rangle|y\rangle \rightarrow |x\rangle|y \oplus E_k(x)\rangle$. Consequently, a vast array of classical symmetric cryptosystems have already been proven fundamentally devastated within the Q2 model, with catastrophic, polynomial-time key recovery attacks routinely achieved by leveraging Simon’s period-finding algorithms or Grover’s amplitude amplification [12, 13].

In principle, the logic circuit of any fully classical computational function f can be methodically transformed into a reversible quantum circuit U_f . Therefore, if the algorithm’s circuit is entirely public and keyless, operating under the assumption that an attacker can evaluate U_f via superposition is a mathematically sound cryptographic premise. However, this foundational assumption alters fundamentally when secret symmetric keys are structurally integrated. If f is defined as a key-dependent cryptographic function E_k , an external network attacker can only realistically gain quantum superposition access to it if the honest party deliberately and explicitly engineers a quantum interface. While the Q2 model undeniably holds significant theoretical value—particularly in highly esoteric contexts like cryptographic circuit obfuscation, theoretical white-box cryptography, or specific instantiations of delegated quantum computation—it represents an overly pessimistic, highly improbable threat model for standard symmetric encryption deployments. In the vast majority of real-world operational deployments, honest parties (e.g., standard enterprise servers, TLS endpoints, or certified Hardware Security Modules) process cryptographic primitives exclusively via isolated classical hardware, strictly accepting discrete classical inputs and returning discrete classical outputs. Even if these systems were to be eventually deployed on advanced quantum architecture in the future, there is absolutely no practical, functional, or economic incentive to expose a quantum query interface for a symmetric cipher to a potentially untrusted external network.

Therefore, the **Q1 model** aligns far more accurately with actual “Harvest Now, Decrypt Later” threat landscapes and highly realistic deployment scenarios. Within this restricted model, the attacker is authorized to leverage unbounded quantum superposition queries against public, keyless components (such as standardized public permutations or unkeyed hash functions) but remains strictly confined to classical, sequential point-wise queries when actively interacting with the keyed construction E_k . Given the absolute unavailability of a dedicated, interactive quantum oracle, the attacker cannot synthesize an effective quantum query circuit U_{E_k} for a specific cryptographic instance initialized with an unknown, randomly sampled key k . Consequently, establishing irrefutable security bounds under the Q1 model directly determines how critical real-world systems relying on symmetric cryptographic primitives should be selected, securely deployed, and adequately parameterized in the post-quantum computing era.

From a perspective strictly grounded in provable security, the Q1 model introduces uniquely intractable analytical and mathematical intricacies that fundamentally diverge from those encountered in the Q2 model. In the unrestricted Q2 setting, the homogeneity of the adversary’s quantum access easily exposes classical symmetric designs to devastating algorithmic attacks, routinely rendering their original classical security proofs completely invalid and necessitating total structural redesigns. Conversely, while the adversary’s operational capabilities are undeniably restricted in the realistic Q1 model, the core mathematical difficulty for executing security reductions stems entirely from the **hybrid, fundamentally asymmetric access paradigm**. Specifically, the adversary can issue unbounded, highly entangled quantum queries to thoroughly explore the global algebraic state of the public permutations, yet is simultaneously confined to extracting highly sparse, discrete classical information from the

keyed cipher. This operational dichotomy strictly prevents the direct, naive application of standard quantum cryptanalysis algorithms, forcing complex security proofs to rigorously and meticulously manage the adaptive, highly correlated interdependencies between expansive quantum amplitudes and discrete classical query transcripts. Accurately bounding the statistical inconsistencies an adversary might cunningly exploit through these interleaved, cross-domain queries demands the deployment of highly fine-grained analytical tools, most notably adaptive reprogramming and resampling lemmas. Crucially, proof techniques inherently tailored for the restrictive Q1 setting cannot simply be derived by trivially downgrading Q2 methodologies, as any mathematical attempt by the reduction algorithm to explicitly track or isolate an adversary’s specific quantum queried coordinates invariably triggers an irreversible quantum measurement collapse [3]. This highlights the profound methodological necessity of transcript-based reduction techniques entirely unique to the Q1 theoretical framework.

In this vital research direction, the global cryptographic community has achieved a series of pivotal breakthroughs, laying an exceptionally solid mathematical foundation for Q1 security analysis. Jaeger et al. [14] executed foundational work by proving a Q1 security lower bound for the FX construction, successfully demonstrating it guarantees robust $(k + n)/3$ -bit security under non-adaptive attacks, which tightly matches the cryptanalytic upper bounds previously achieved by Bonnetain et al. [15]. Alagic et al. [16] subsequently achieved a monumental landmark result by rigorously proving that the fundamental Even-Mansour construction inherently ensures $n/3$ -bit post-quantum Q1 security under fully adaptive attacks, establishing a completely tight bound against the highly optimized offline Simon’s algorithm attack. Expanding iteratively on this framework, Alagic et al. [17] proved the security of the Tweakable Even-Mansour construction under the Q1 model, effectively preserving the critical $n/3$ -bit security guarantee. Advancing the field further, Guo et al. [18] successfully extended the security proof of the FX construction directly into highly aggressive adaptive chosen-plaintext-and-ciphertext attack scenarios. Recently, comprehensive research by Bai et al. [19] systematically investigated the entire class of Key-Alternating Ciphers (t-KAC), mathematically proving robust Q1 security boundaries for both foundational 2-round and generalized t -round variant architectures.

However, despite these extensive, field-defining advancements, a rigorous mathematical evaluation and theoretical formalization of the composite SoEM construction [8] strictly under the Q1 model has remained critically unaddressed. While Shinagawa et al. [20] previously demonstrated a devastating polynomial-time key recovery attack against SoEM21 in the strongly adversarial Q2 model by directly leveraging Simon’s algorithm, recent state-of-the-art cryptanalysis conducted by Li et al. [21] has decisively proven that in the highly realistic Q1 model, executing an attack utilizing an offline adaptation of Simon’s algorithm strictly necessitates an asymptotic complexity of $\tilde{O}(2^{n/3})$. While this recent discovery successfully establishes a clear and undisputed upper bound for attack efficiency within the model, a mathematically rigorous, provable security lower bound explicitly designed to match this complexity and close the theoretical gap has been notably absent from current literature. In resolving this open problem, the proof techniques developed in our work have already demonstrated significant methodological generality. Notably, recent research by Datta

et al. [22] evaluating the Keyed Sum of Permutations (KSoP) construction and its variants explicitly adopted the hybrid sequence and transcript-driven reprogramming techniques established in our framework. Their successful application of our approach to establish matching Q1 bounds for alternative summation-type symmetric primitives provides strong external validation of our methodology’s broad applicability.

Construction	Threat Model	Upper Bound (Known Attack)	Lower Bound (Provable Security)
SoEM21	Classical	$O(2^{n/2})$ [8]	$\Omega(2^{n/2})$ [8]
	Q1	$\tilde{O}(2^{n/3})$ [21]	$\Omega(2^{n/3})$ [Theorem 3]
	Q2	Polynomial $O(n)$ [20]	–
Even-Mansour	Classical	$O(2^{n/2})$ [23]	$\Omega(2^{n/2})$ [23]
	Q1	$\tilde{O}(2^{n/3})$ [15]	$\Omega(2^{n/3})$ [16]
	Q2	Polynomial $O(n)$ [24]	Polynomial $O(n)$ [25]
FX Construction	Classical	$O(2^{(n+k)/2})$ [26]	$\Omega(2^{(n+k)/2})$ [26]
	Q1	$\tilde{O}(2^{(n+k)/3})$ [15]	$\Omega(2^{(k+n)/3})$ [14]
	Q2	$O(n2^{k/2})$ [27]	–
Tweakable EM	Classical	$O(2^{n/2})$ [28]	$\Omega(2^{n/2})$ [28]
	Q1	$\tilde{O}(2^{n/3})$ [15]	$\Omega(2^{n/3})$ [17]
	Q2	Polynomial $O(n)$ [24]	Polynomial $O(n)$ [25]
t-KAC	Classical	$O(2^{\frac{t}{t+1}n})$ [29]	$\Omega(2^{\frac{t}{t+1}n})$ [30]
	Q1	$O(2^{\frac{t(t+1)n}{(t+1)^2+1}})$ [19]	$\Omega(2^{\frac{tn}{2t+1}})$ [19]
	Q2	Polynomial $O(n)$ [31]	–

Table 1 Comprehensive post-quantum security bounds for prominent symmetric cryptographic constructions. The core theoretical contribution of this paper is explicitly highlighted in bold, successfully establishing the tight lower bound for SoEM21 specifically within the Q1 adversarial model.

1.1 Contributions of This Paper

As established in the preceding analysis, while the optimal cryptanalytic attack complexity against the SoEM21 variant in the Q1 model is firmly known to be $\tilde{O}(2^{n/3})$ [21], a rigorous, mathematical provable security lower bound required to formalize this limit and verify its absolute tightness remains conspicuously absent. To bridge this critical theoretical gap and provide a holistic security assessment, this paper conducts an exhaustive and systematic reduction of the SoEM21 construction, directly aiming to construct an irrefutable Q1 security foundation. The primary contributions of this comprehensive paper are rigorously summarized as follows:

1. Establishing a Complete and Tight Q1 Security Bound for SoEM21:

The paramount theoretical contribution of this work is the first mathematically rigorous proof that the SoEM21 construction intrinsically guarantees a robust $n/3$ -bit security floor strictly under the Q1 quantum adversarial model. By systematically and formally verifying the construction’s exponential resistance to transcript-based, adaptive quantum attacks, our mathematically proven security lower bound exactly matches the $\tilde{O}(2^{n/3})$ upper bound established by the most optimized known attacks currently available in literature [21]. This unification

effectively merges the cryptanalytic and provable security perspectives, decisively confirming that no theoretical Q1 adversary can breach this specific complexity threshold, thereby supplying a definitive, mathematically sound security assurance for the real-world operational deployment of SoEM-based cryptographic schemes.

2. **Advancing Quantum Lower-Bound Proof Techniques for Composite Permutations:** Unlike basic, single-permutation architectures such as the standard Even-Mansour cipher, the SoEM21 construction fundamentally utilizes the parallel invocation of two entirely independent permutations intimately coupled with the modular superposition of a single, unified key. This structural bifurcation and parallel execution exponentially increase the mathematical complexity required for effectively managing adaptive interdependencies between interleaved quantum queries. By engineering a novel, highly sophisticated combined function logic and innovatively deploying the **Reprogramming Lemma** seamlessly alongside the **Resampling Lemma**, this paper successfully tames the severe statistical perturbations caused by an adversary’s highly correlated, interleaved quantum queries across multiple public permutations. This rigorous proof framework demonstrates significant architectural generality and provides a vital, reusable methodological reference for analyzing multi-branch or complex summation-based cryptographic components in future research endeavors.
3. **Clarifying the Structural Cryptographic Resilience of the SoEM Paradigm:** Our extensive theoretical analysis conclusively demonstrates that the Q1 post-quantum security guarantees of SoEM21 do not inherently depend on the ultimate secrecy or the obfuscated internal structure of the underlying public permutations. Instead, the security is intrinsically guaranteed by its unique key superposition logic and robust modular algebraic structure. This fundamental “permutation-independent” resilience characteristic significantly bolsters the viability of permutation-based constructions in ongoing post-quantum cryptography standardization efforts initiated by bodies like NIST. Furthermore, it provides direct, foundational theoretical support for evaluating the quantum resistance of complex downstream protocols built explicitly upon the SoEM design philosophy, such as the encryption scheme CENCPP* and the message authentication code nEHTm.

2 Preliminaries

2.1 Notation and Model

Let n be the security parameter. All operations in this paper are considered over $\{0, 1\}^n$. Let $\mathcal{P}_n$ denote the set of all permutations from $\{0, 1\}^n$ to $\{0, 1\}^n$. We uniformly and independently sample public permutations P_1 and P_2 from $\mathcal{P}_n$. The key k is a bit string uniformly sampled from $\{0, 1\}^n$.

SoEM21 Construction: Defined as a function $E_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$, computed as follows:

$$E_k(x) = P_1(x \oplus k) \oplus P_2(x \oplus k) \oplus k.$$

The specific architecture and data flow of the SoEM21 variant are illustrated in detail (See Fig. 1). The visual model highlights the initial bifurcation of the plaintext, the parallel invocation of the two independent public permutations, and the uniform application of the single cryptographic key across all branches.

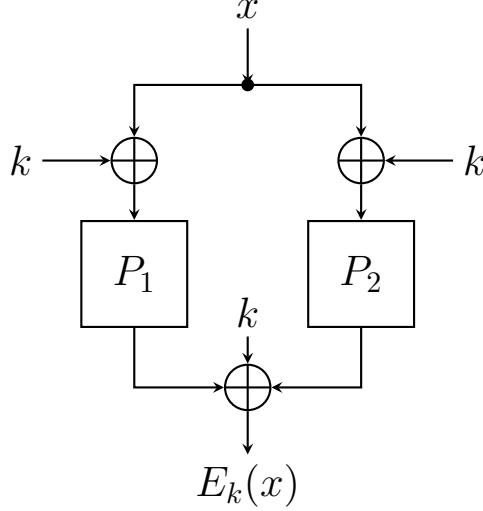


Fig. 1 The SoEM21 cipher.

We consider the adversarial capability of an adversary $\mathcal{A}$ in the **Q1 model**: $\mathcal{A}$ can make **quantum queries** to the public permutations P_1 and P_2 (and their inverses). This means $\mathcal{A}$ can apply the unitary operator $U_P : |x\rangle|y\rangle \rightarrow |x\rangle|y \oplus P(x)\rangle$ and its inverse. $\mathcal{A}$ can only make **classical queries** to the keyed construction E_k . That is, $\mathcal{A}$ provides a classical input x and receives the classical output $E_k(x)$. We denote the number of classical queries made by adversary $\mathcal{A}$ as q_E , and the total number of quantum queries (the sum of queries to P_1 and P_2) as q_P .

2.2 Security Definition

Our goal is to prove that SoEM21 is a Pseudorandom Function (PRF). We formalize its security through a distinguishing game. Unlike Pseudorandom Permutations (PRPs), PRFs are not required to be invertible. Consequently, when proving the security of a PRF, the reduction game is typically designed to only allow forward queries. This is because the security definition of a PRF focuses on the indistinguishability of the function's output from a random function, without considering the computability of the inverse operation. Allowing inverse queries would introduce capabilities that do not align with the PRF security goal and could invalidate the reduction. Therefore, we assume that the adversary is limited to forward queries of the form $E_k(x) = P_1(x \oplus k) \oplus P_2(x \oplus k) \oplus k$.

Definition 1 (PRF Security Game for SoEM21 in the Q1 Model) The game involves a challenger and a quantum adversary $\mathcal{A}$. The challenger flips a uniform random bit b .

- If $b = 0$ (**Real World**): The challenger randomly selects a key $k \leftarrow \{0, 1\}^n$ and permutations $P_1, P_2 \leftarrow \mathcal{P}_n$. The adversary $\mathcal{A}$ is granted **quantum query access** to $P_1, P_2, P_1^{-1}, P_2^{-1}$ and **classical query access** to $E_k(\cdot) = P_1(\cdot \oplus k) \oplus P_2(\cdot \oplus k) \oplus k$.
- If $b = 1$ (**Ideal World**): The challenger samples an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with the same domain and range as E_k , and permutations $P_1, P_2 \leftarrow \mathcal{P}_n$. The adversary $\mathcal{A}$ is granted **quantum query access** to $P_1, P_2, P_1^{-1}, P_2^{-1}$ and **classical query access** to $R(\cdot)$.

At the end of the game, the adversary $\mathcal{A}$ outputs a guess bit b' . The advantage of adversary $\mathcal{A}$ in this game is defined as:

$$\text{Adv}_{\text{SoEM21}}^{\text{Q1-PRF}}(\mathcal{A}) = |\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]|.$$

More generally, for any adversary making at most q_E classical queries and q_P quantum queries, we define:

$$\text{Adv}_{\text{SoEM21}}^{\text{Q1-PRF}}(q_E, q_P) = \max_{\mathcal{A}} \text{Adv}_{\text{SoEM21}}^{\text{Q1-PRF}}(\mathcal{A}).$$

We now introduce Alagic et al.'s Reprogramming Lemma and Resampling Lemma [16, 17].

2.3 Reprogramming Lemma

The Reprogramming Lemma applies to the following experiment: A distinguisher D chooses an arbitrary function F and a randomized procedure B that determines a set B of points where F may be reprogrammed to some known value. Then, D is given quantum access to either F or a reprogrammed version of F ; after it finishes its oracle queries, D is given B . The lemma states that as long as no point is reprogrammed with high probability, D cannot determine whether it interacted with F or its reprogrammed version.

For a function $F : \{0, 1\}^m \rightarrow \{0, 1\}^n$ and a set $B \subset \{0, 1\}^m \times \{0, 1\}^n$ such that each $x \in \{0, 1\}^m$ is the first element of at most one tuple in B , define

$$F^{(B)}(x) = \begin{cases} y & \text{if } (x, y) \in B \\ F(x) & \text{otherwise} \end{cases}.$$

Lemma 1 (Reprogramming Lemma) [16] *Let D be a quantum distinguisher in the following experiment:*

Phase 1: D outputs a description of a function $F_0 = F : \{0, 1\}^m \rightarrow \{0, 1\}^n$ and a description of a randomized algorithm B whose output is a set $B \subset \{0, 1\}^m \times \{0, 1\}^n$, where each $x \in \{0, 1\}^m$ is the first element of at most one tuple in B . Let $B_1 = \{x \mid \exists y : (x, y) \in B\}$ and

$$\epsilon = \max_{x \in \{0, 1\}^m} \{\Pr[x \in B_1]\}.$$

Phase 2: Run B to obtain B . Let $F_1 = F^{(B)}$. Choose a uniform random bit b , and D is given quantum access to F_b .

Phase 3: D loses access to F_b and receives the randomness r used to invoke B in Phase 2. Then D outputs a guess b' .

For any D that makes at most q queries to its oracle in expectation,

$$|\Pr[D \text{ outputs } 1 \mid b = 1] - \Pr[D \text{ outputs } 1 \mid b = 0]| \leq 2q \cdot \sqrt{\epsilon}.$$

2.4 Resampling Lemma

The Resampling Lemma applies to the following experiment: First, a distinguisher D is given quantum access to a random permutation P . Then, in a second phase, P may be reprogrammed so that its value at a single uniform point s is changed to an independent uniform value. Since the distribution of $P(s)$ is identical before and after any such reprogramming, we call it “resampling”. D ’s goal is to determine whether its oracle was resampled. That is, D can tell only if it placed significant amplitude on s in a query during the first phase, even if given s and continued oracle access in the second phase.

Given a permutation $P : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and $s, y \in \{0, 1\}^n$, define the reprogrammed function $P_{s \rightarrow y} : \{0, 1\}^n \rightarrow \{0, 1\}^n$ as

$$P_{s \rightarrow y}(w) = \begin{cases} y & \text{if } w = s \\ P(w) & \text{otherwise} \end{cases}.$$

Lemma 2 (Resampling Lemma) [16] *Let D be a quantum distinguisher in the following two-phase experiment:*

Phase 1: D is given quantum access to a uniform random permutation $P_0 = P$ and its inverse $P_0^{-1} = P^{-1}$.

Phase 2: Uniformly choose $s_0, s_1 \in \{0, 1\}^n$. Define P_1 : if $b = 0$, then $P_1 = P$; if $b = 1$, then $P_1 = P \circ \text{swap}_{s_0, s_1}$, where swap_{s_0, s_1} swaps the values at s_0 and s_1 . D receives s_0, s_1 and is given quantum access to P_1, P_1^{-1} . Finally, D outputs a guess bit b' .

For D making at most q queries to P_0, P_0^{-1} in Phase 1, we have:

$$|\Pr[D \text{ outputs } b' = 0] - \Pr[D \text{ outputs } b' = 1]| \leq 4\sqrt{\frac{q}{2^n}}.$$

3 Post-Quantum Security of the SoEM21 Construction

We now establish the post-quantum security of the SoEM21 construction based on the lemmas from the previous section. Recall that the SoEM21 construction is defined as: $E_k(x) = P_1(x \oplus k) \oplus P_2(x \oplus k) \oplus k$, where $P_1, P_2 : \{0, 1\}^n \rightarrow \{0, 1\}^n$ are independent public random permutations, and $k \in \{0, 1\}^n$ is a uniform key. Our proof considers an adversary $\mathcal{A}$ who can make classical queries to E_k (and its inverse, if applicable) and quantum queries to P_1, P_2 (and their inverses). $\mathcal{A}$ ’s goal is to distinguish the real world (interacting with $E_k[P_1, P_2]$ and P_1, P_2) from the ideal world (interacting with an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and independent

random permutations P_1, P_2). In the following, we let $\mathcal{P}_n$ denote the set of all permutations over $\{0, 1\}^n$. We write $E_k[P_1, P_2]$ to emphasize the dependence on P_1, P_2 . Our main result is as follows:

Theorem 3 (Main Theorem) *For any adversary $\mathcal{A}$ in the Q1 model making q_E classical queries and q_P quantum queries, it holds that:*

$$\text{Adv}_{\text{SoEM21}}^{Q1\text{-PRF}}(q_E, q_P) \leq 12 \cdot \frac{q_E \sqrt{q_P} + q_P \sqrt{q_E}}{2^{n/2}}.$$

In particular, when both q_E and q_P are far less than $2^{n/3}$, the adversary's advantage is negligible. This proves that SoEM21 has $n/3$ -bit security under the Q1 model.

3.1 Formal Proof of the Main Theorem

Proof of Theorem 3 We now utilize the established quantum lemmas to formally construct the security reduction for the SoEM21 architecture. The entire execution timeline of the quantum adversary $\mathcal{A}$ is mathematically partitioned into $q_E + 1$ distinct computational phases, sequentially labeled $0, \dots, q_E$. Phase j corresponds strictly to the temporal interval existing between the adversary's j -th and $(j + 1)$ -th classical queries to the respective oracles. The adversary is explicitly permitted to adaptively and arbitrarily distribute its total quantum query budget q_P across these sequential phases, maximizing its potential distinguishing advantage. We define $q_{P,j}$ as the expected number of quantum superposition queries executed during phase j , maintaining the strict mathematical condition that:

$$\sum_{j=0}^{q_E} q_{P,j} = q_P. \quad (1)$$

Let the ordered historical transcript of the adversary's first j classical queries be denoted as the set $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$. To maintain absolute algebraic consistency between this expanding classical transcript and the underlying quantum oracles, we must dynamically reprogram the public permutations in a highly controlled manner. For each recorded query index $i \in \{1, \dots, j\}$, we define the key-dependent input coordinate as $a_i = x_i \oplus k$. We then sequentially sample uniformly random intermediate values $u_i \in \{0, 1\}^n$ from the global state space.

To successfully reprogram the primary permutation P_1 without triggering measurement collapse, we utilize quantum-secure swap operations. If the current evaluation $P_1(a_i) \neq u_i$, we locate the unique inverse coordinate z_i such that $P_1(z_i) = u_i$ and execute a discrete swap at these specific coordinates. This mathematical operation guarantees that the updated permutation satisfies $P_{1,T_j,k}(a_i) = u_i$. We represent this cumulative, sequential swap logic as the combined unitary operator:

$$S_{T_j, P_1, k} = \prod_{i=1}^j \text{swap}_{a_i, z_i}, \quad S_{T_j, P_1, k} \circ P_1 = P_{1, T_j, k}.$$

Concurrently, to reprogram the secondary parallel permutation P_2 , we calculate the required target output state $t_i = k \oplus u_i \oplus y_i$. If $P_2(a_i) \neq t_i$, we identify the coordinate w_i such that $P_2(w_i) = t_i$ and execute an identical swap protocol, strictly ensuring $P_{2, T_j, k}(a_i) = t_i$. At this point we have:

$$S_{T_j, P_2, k} = \prod_{i=1}^j \text{swap}_{a_i, w_i}, \quad S_{T_j, P_2, k} \circ P_2 = P_{2, T_j, k}.$$

$$P_{1,T_j,k}(x_i \oplus k) = u_i, \quad P_{2,T_j,k}(x_i \oplus k) = u_i \oplus k \oplus y_i.$$

This dual-reprogramming mechanism yields the vital algebraic property that perfectly simulates the expected classical cipher output without introducing measurable transcript collisions:

$$P_{1,T_j,k}(x_i \oplus k) \oplus P_{2,T_j,k}(x_i \oplus k) \oplus k = y_i. \quad (2)$$

With this transcript consistency mechanism rigorously established, we formally define the baseline hybrid experiments H_j and H'_j to progressively orchestrate the mathematical transition from the real-world cryptographic deployment to the ideal statistical simulation.

We now define hybrid experiments H_j for $j = 0, 1, \dots, q_E$.

Experiment H_j :

1. The distinguisher $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. It then runs $\mathcal{A}$, answering its quantum queries using quantum oracle access to P_1 and P_2 , and answering its classical queries using oracle access to R , **stopping immediately before its $(j + 1)$ -st classical query**. Let $\mathbf{T}_j = \{(\mathbf{x}_1, \mathbf{y}_1), \dots, (\mathbf{x}_j, \mathbf{y}_j)\}$ be the list of classical query-response pairs.
3. For the remaining execution of $\mathcal{A}$, use $E_k[P_{1,T_j,k}, P_{2,T_j,k}]$ to answer its classical queries, and use $P_{1,T_j,k}, P_{2,T_j,k}$ to answer its quantum queries.

We now define hybrid experiments H'_j for $j = 0, 1, \dots, q_E - 1$.

Experiment H'_j :

1. $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. It then runs $\mathcal{A}$, answering its quantum queries using quantum oracle access to P_1 and P_2 , and answering its classical queries using oracle access to R , **stopping immediately after its $(j + 1)$ -st classical query**. Let $\mathbf{T}_{j+1} = \{(\mathbf{x}_1, \mathbf{y}_1), \dots, (\mathbf{x}_{j+1}, \mathbf{y}_{j+1})\}$ be the list of classical query-response pairs.
3. For the remaining execution of $\mathcal{A}$, use $E_k[P_{1,T_{j+1},k}, P_{2,T_{j+1},k}]$ to answer its classical queries, and use $P_{1,T_{j+1},k}, P_{2,T_{j+1},k}$ to answer its quantum queries.

To intuitively illustrate the distinction between the experiments, their respective oracle sequences are depicted as follows:

Oracle sequence for H_j :

$$\underbrace{(P_1, P_2, R)^j}_{j \text{ classical queries}}, \quad \underbrace{(P_{1,T_j,k}, P_{2,T_j,k}, E_k[P_{1,T_j,k}, P_{2,T_j,k}])^{q_E-j}}_{q_E-j \text{ remaining classical queries}}$$

Oracle sequence for H'_j :

$$\underbrace{(P_1, P_2, R)^{j+1}}_{j+1 \text{ classical queries}}, \quad \underbrace{(P_{1,T_{j+1},k}, P_{2,T_{j+1},k}, E_k[P_{1,T_{j+1},k}, P_{2,T_{j+1},k}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining classical queries}}$$

In subsequent Lemma 4 and Lemma 5, we establish bounds on the distinguishability between H_j and H'_j and between H'_j and H_{j+1} . For $0 \leq j < q_E$ we have:

$$|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| \leq 8\sqrt{\frac{q_P}{2^n}} + \frac{3j}{2^n} + \frac{2q_E}{2^n}, \quad (3)$$

$$|\Pr[\mathcal{A}(H'_j) = 1] - \Pr[\mathcal{A}(H_{j+1}) = 1]| \leq 2 \cdot q_{P,j+1} \sqrt{2(j+1)/2^n}. \quad (4)$$

Then we have:

$$\begin{aligned} & |\Pr[\mathcal{A}(H_0) = 1] - \Pr[\mathcal{A}(H_{q_E}) = 1]| \\ & \leq \sum_{j=0}^{q_E-1} [|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| + |\Pr[\mathcal{A}(H'_j) = 1] - \Pr[\mathcal{A}(H_{j+1}) = 1]|] \\ & \leq \sum_{j=0}^{q_E-1} \left[8\sqrt{\frac{q_P}{2^n}} + \frac{3j}{2^n} + \frac{2q_E}{2^n} + 2 \cdot q_{P,j+1} \sqrt{\frac{2(j+1)}{2^n}} \right] \\ & \leq \sum_{j=0}^{q_E-1} \left(8\sqrt{\frac{q_P}{2^n}} + \frac{3j}{2^n} + \frac{2q_E}{2^n} + 2 \cdot q_{P,j+1} \sqrt{\frac{2(j+1)}{2^n}} \right) \\ & \leq 2 \cdot q_E \cdot (q_E - 1) \cdot 2^{-n} + \frac{3 \cdot q_E \cdot (q_E - 1)}{2} \cdot 2^{-n} + \sum_{j=0}^{q_E-1} \left(8\sqrt{\frac{q_P}{2^n}} + 2 \cdot q_{P,j+1} \sqrt{\frac{2q_E}{2^n}} \right) \end{aligned}$$

Next, we proceed to further simplify the bound obtained. Note that if $q_P = 0$, the adversary makes no quantum queries, and hence E_k and R are perfectly indistinguishable; in this case, the theorem holds trivially. Therefore, without loss of generality, we assume $q_P \geq 1$ in the following. Therefore, we can assume $q_P \geq 1$. We can also assume $q_E < 2^{n/2}$, otherwise the bound would exceed 1. Under these assumptions, we have $2^{-n} q_E^2 \leq 2^{-n/2} q_E$ and $2^{-n/2} q_E \leq 2^{-n/2} q_E \sqrt{q_P}$. Thus,

$$\begin{aligned} |\Pr[\mathcal{A}(H_0) = 1] - \Pr[\mathcal{A}(H_{q_E}) = 1]| & \leq 4 \cdot q_E \sqrt{q_P} \cdot 2^{-n/2} + 2^{-n/2} (8q_E \sqrt{q_P} + 3q_P \sqrt{q_E}) \\ & \leq 12 \cdot 2^{-n/2} (q_P \sqrt{q_E} + q_E \sqrt{q_P}). \end{aligned}$$

A constant-level advantage is achieved when $2^{-n/2} (q_P \sqrt{q_E} + q_E \sqrt{q_P}) \approx 1$, implying $\Omega(2^{n/3})$ queries are required. $\square$

3.2 Proof of Intermediate Quantum Lemmas

To formally validate the overarching bounds utilized in the summation above, we now present the rigorous proofs governing the statistical perturbations of the intermediate state transitions.

Lemma 4 *Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E - 1$, we have:*

$$|\Pr[\mathcal{A}(H'_j) = 1] - \Pr[\mathcal{A}(H_{j+1}) = 1]| \leq 2 \cdot q_{P,j+1} \sqrt{2(j+1)/2^n}.$$

Proof We can write the oracle sequences defined by H'_j and H_{j+1} as follows:

$$\begin{aligned} H'_j : & \underbrace{(P_1, P_2, R)^{j+1}}_{j+1 \text{ queries}}, \underbrace{\mathbf{P}_1, \mathbf{T}_{j+1}, \mathbf{k}, \mathbf{P}_2, \mathbf{T}_{j+1}, \mathbf{k}}_{\text{phase } j+1 \text{ quantum}}, \underbrace{(P_{1,T_{j+1},k}, P_{2,T_{j+1},k}, E_k[P_{1,T_{j+1},k}, P_{2,T_{j+1},k}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining queries}} \\ H_{j+1} : & \underbrace{(P_1, P_2, R)^{j+1}}_{j+1 \text{ queries}}, \underbrace{\mathbf{P}_1, \mathbf{P}_2}_{\text{phase } j+1 \text{ quantum}}, \underbrace{(P_{1,T_{j+1},k}, P_{2,T_{j+1},k}, E_k[P_{1,T_{j+1},k}, P_{2,T_{j+1},k}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining queries}} \end{aligned}$$

We now define a combined function to facilitate the use of the [Reprogramming Lemma](#). Let $F : \{1, 2\} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$, where $F(1, x) = P_1(x)$, corresponding to queries to P_1 , and $F(2, x) = P_2(x)$, corresponding to queries to P_2 . The adversary's quantum queries to P_1 and P_2 are now viewed as queries to F . Therefore, the adversary's total quantum queries to F is q_P (the sum of queries to P_1 and P_2).

Let $\mathcal{A}$ be a distinguisher between H'_j and H_{j+1} . We construct a distinguisher $\mathcal{D}$ for the blinding experiment in Lemma 1:

Phase 1: $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$. It then runs $\mathcal{A}$, using P_1, P_2 to answer its quantum queries and using R to answer classical queries, until responding to $\mathcal{A}$'s $(j+1)$ -th classical query. Let $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$ be the list of classical query-response pairs. $\mathcal{D}$ defines the combined function $F : \{1, 2\} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ as described above. $\mathcal{D}$ implicitly defines a reprogrammed version of F , denoted $F^{(B)}$, such that for all $a \in \{1, 2\}$ and $x \in \{0, 1\}^n$, we have $F^{(B)}(a, x) = P_{a, T_{j+1}, k}(x)$.

Phase 2: $\mathcal{D}$ generates the reprogrammed set B based on the $(j+1)$ -th query, and is given quantum access to an oracle F_b . $\mathcal{D}$ continues running $\mathcal{A}$, answering its quantum queries via $P_a = F_b(a, \cdot)$. When $\mathcal{A}$ issues its next (i.e., the $(j+2)$ -th) classical query, Phase 2 ends.

Phase 3: $\mathcal{D}$ uses B to generate k . It continues running $\mathcal{A}$, using E_k to answer its classical queries and using $P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k}$ to answer its quantum queries. Finally, it outputs whatever $\mathcal{A}$ outputs.

Note that $\mathcal{D}$ is a valid distinguisher for the reprogramming experiment in Lemma 1. If $b = 0$ (i.e., the oracle of $\mathcal{D}$ in Phase 2 is the original $F_0 = F$), then the output of $\mathcal{A}$ is distributed identically to its output in H_{j+1} . If $b = 1$ (i.e., the oracle of $\mathcal{D}$ in Phase 2 is the reprogrammed $F_1 = F^{(B)}$), then the output of $\mathcal{A}$ is distributed identically to its output in H'_j . Therefore, $|\Pr[\mathcal{A}(H'_j) = 1] - \Pr[\mathcal{A}(H_{j+1}) = 1]|$ exactly equals the distinguishing advantage of $\mathcal{D}$.

To bound this advantage using Lemma 1, we must limit the expected number of queries $\mathcal{D}$ makes in Phase 2 (when $F = F_0$) and the maximum reprogramming probability ϵ :

Bounding Phase 2 Queries: When $F = F_0$, the expected number of queries $\mathcal{D}$ makes in Phase 2 equals the expected number of queries $\mathcal{A}$ makes in phase $(j+1)$ of H_{j+1} . Since H_{j+1} and H'_j are identical up to the completion of phase $(j+1)$, this is precisely $q_{P, j+1}$.

Bounding Reprogramming Probability ϵ : To strictly preserve the bijectivity of the permutations without creating output collisions, the reprogramming of P_1 and P_2 at the key-dependent operational point $a_i = x_i \oplus k$ is implemented via swap operations. For any query $i \in \{1, \dots, j+1\}$, choosing a uniformly random intermediate value $u_i \in \{0, 1\}^n$ means: 1. Reprogramming $P_1(a_i)$ to u_i implicitly swaps the outputs at a_i and the inverse coordinate $P_1^{-1}(u_i)$. 2. Reprogramming $P_2(a_i)$ to $k \oplus u_i \oplus y_i$ implicitly swaps the outputs at a_i and the inverse coordinate $P_2^{-1}(k \oplus u_i \oplus y_i)$.

Therefore, the set of reprogrammed inputs for F is a subset of $B_1 = \bigcup_{i=1}^{j+1} \{(1, a_i), (1, P_1^{-1}(u_i)), (2, a_i), (2, P_2^{-1}(k \oplus u_i \oplus y_i))\}$. For any fixed coordinate $(s, x) \in \{1, 2\} \times \{0, 1\}^n$, we calculate the probability it is reprogrammed by the i -th query. Because the hidden key k is uniformly random and independent of x_i , the forward coordinate a_i is perfectly uniform. Similarly, because the intermediate mask u_i is uniformly sampled, the inverse coordinates are also uniformly distributed.

Applying the union bound over these mutually exclusive targets for a single query yields $\Pr[\text{point is reprogrammed by the } i\text{-th query}] \leq \frac{1}{2^n} + \frac{1}{2^n} = \frac{2}{2^n}$. To capture the cumulative risk over the entire historical transcript of $j+1$ classical queries, we apply a final union bound

over the summation index i :

$$\epsilon = \max_{(s,x)} \Pr[(s,x) \in B_1] \leq \sum_{i=1}^{j+1} \frac{2}{2^n} = \frac{2(j+1)}{2^n}$$

Applying Lemma 1 with these bounds, we obtain:

$$|\Pr[\mathcal{A}(H'_j) = 1] - \Pr[\mathcal{A}(H_{j+1}) = 1]| \leq 2 \cdot q_{P,j+1} \sqrt{\frac{2(j+1)}{2^n}}.$$

□

Lemma 5 *Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E$, we have:*

$$|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| \leq 8\sqrt{\frac{q_P}{2^n}} + \frac{3j}{2^n} + \frac{2q_E}{2^n}.$$

Proof To prove Lemma 5, we introduce auxiliary experiments: H_j^* , H_j^{**} , H_j^{***} .

(Note: These intermediate hybrids progressively decouple the key k from the adversary's prior quantum observations via permutation resampling, bridging the algebraic transition to the ideal random function R .)

Experiment H_j^* :

1. $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. Run $\mathcal{A}$, using R to answer its classical queries and P_1, P_2 to answer its quantum queries, until $\mathcal{A}$ issues the $(j+1)$ -th classical query x_{j+1} .
3. Uniformly sample $s_0, s_1 \in \{0, 1\}^n$, define $k = s_0 \oplus x_{j+1}$. Define $P_1^{(1)}$ as

$$P_1^{(1)}(x) = P_1 \circ \text{swap}_{s_0, s_1}(x)$$

(i.e., apply swap of s_0 and s_1 to P_1). Then continue running $\mathcal{A}$, using E_k to answer its remaining classical queries (including the $(j+1)$ -th), and using $P_{1,T_j,k}^{(1)}, P_{2,T_j,k}$ to answer its quantum queries, where $P_{1,T_j,k}^{(1)} = S_{T_j, P_1, k} \circ P_1^{(1)}$.

Experiment H_j^{} :**

1. $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. Run $\mathcal{A}$, using R to answer its classical queries and P_1, P_2 to answer its quantum queries, until $\mathcal{A}$ issues the $(j+1)$ -th classical query x_{j+1} .
3. Uniformly sample $s_0, s_1 \in \{0, 1\}^n$, define $k = s_0 \oplus x_{j+1}$. Define $P_1^{(1)}$ as

$$P_1^{(1)}(x) = P_1 \circ \text{swap}_{s_0, s_1}(x)$$

and define $P_2^{(1)}$ as

$$P_2^{(1)}(x) = P_2 \circ \text{swap}_{s_0, s_1}(x)$$

. Then continue running $\mathcal{A}$, using E_k to answer its remaining classical queries (including the $(j+1)$ -th), and using $P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}$ to answer its quantum queries, where $P_{1,T_j,k}^{(1)} = S_{T_j, P_1, k} \circ P_1^{(1)}$ and similarly for $P_{2,T_j,k}^{(1)}$.

Experiment H_j^{*} :**

1. $\mathcal{D}$ samples two independent random permutations $P_1, P_2 \leftarrow \mathcal{P}_n$ and an independent uniformly random function $R : \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. Run $\mathcal{A}$, using R to answer its classical queries and P_1, P_2 to answer its quantum queries, until the conclusion of $\mathcal{A}$'s $(j+1)$ -th classical query.
3. Uniformly sample $s_0, s_1 \in \{0, 1\}^n$, define $k = s_0 \oplus x_{j+1}$. *Crucial shift:* Use the ideal random function $R(x_{j+1})$ to directly answer the $(j+1)$ -th classical query. Use E_k to answer its remaining classical queries (definitions of $P_{1,T_j,k}^{(1)}$ and $P_{2,T_j,k}^{(1)}$ as above). And use $P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}$ to answer its quantum queries, where $P_{1,T_j,k}^{(1)} = S_{T_j,P_1,k} \circ P_1^{(1)}$ and similarly for $P_{2,T_j,k}^{(1)}$.

To intuitively illustrate the progressive decoupling of the key k in these intermediate experiments, their respective oracle sequences are depicted as follows:

Oracle sequence for H_j^* :

$$\underbrace{(P_1, P_2, R)^j}_{j \text{ queries}}, \underbrace{(P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}, E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}])^{q_E-j}}_{q_E-j \text{ remaining queries}}$$

Oracle sequence for H_j^{**} :

$$\underbrace{(P_1, P_2, R)^j}_{j \text{ queries}}, \underbrace{(P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}, E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}])^{q_E-j}}_{q_E-j \text{ remaining queries}}$$

Oracle sequence for H_j^{***} :

$$\underbrace{(P_1, P_2, R)^j}_{j \text{ queries}}, \underbrace{\mathbf{R}}_{(j+1)\text{-th query}}, \underbrace{(P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}, E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining queries}}$$

We have the following inequalities which will be proven in subsequent lemmas:

$$|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H_j^*) = 1]| \leq 4\sqrt{\frac{q_P}{2^n}}, \quad (5)$$

$$|\Pr[\mathcal{A}(H_j^*) = 1] - \Pr[\mathcal{A}(H_j^{**}) = 1]| \leq 4\sqrt{\frac{q_P}{2^n}}, \quad (6)$$

$$|\Pr[\mathcal{A}(H_j^{**}) = 1] - \Pr[\mathcal{A}(H_j^{***}) = 1]| \leq \frac{j}{2^n}, \quad (7)$$

$$|\Pr[\mathcal{A}(H_j^{***}) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| \leq \frac{2(j + q_E)}{2^n}. \quad (8)$$

Brief justifications for the intermediate bounds:

- **Inequalities (5) and (6)** follow from sequential applications of the Resampling Lemma to P_1 and P_2 , respectively. The adversary's quantum query budget q_P limits its ability to detect the hidden coordinate swap introduced during the resampling phase, leading to a bounded distinguishing advantage of at most $4\sqrt{\frac{q_P}{2^n}}$ for each permutation.
- **Inequality (7)** specifically bounds the distinguishability of the response to the $(j+1)$ -th classical query between the two hybrid experiments. In H_j^{**} , due to the coordinate

swap, the response evaluates to $y_{j+1} = P_{1,T_j,k}(s_1) \oplus P_{2,T_j,k}(s_1) \oplus k$. In H_j^{***} , the response is provided directly by the ideal random function $R(x_{j+1})$. As long as the uniformly random resampling target s_1 does not fall into the set S of j previously queried or reprogrammed points, $P_1(s_1)$ and $P_2(s_1)$ remain unreprogrammed random values, making the algebraic sum uniformly random and indistinguishable from $R(x_{j+1})$. The divergence only occurs if $s_1 \in S$, which would cause the output to collapse to a previous query result y_i , an event bounded by the probability $\frac{j}{2^n}$.

- **Inequality (8)** encapsulates the union bound of “bad events” that would disrupt the strict equality between the reprogrammed permutations. Divergence occurs if the target points hit previous or future queries. Specifically, these events are: Bad_0 where s_1 collides with a past query point in S (probability $\leq \frac{j}{2^n}$); Bad_1 where s_0 collides with a past query point in S , implying a query conflict (probability $\leq \frac{j}{2^n}$); and Bad_2 where any of the q_E subsequent classical queries hit the hidden resampling coordinates s_0 or s_1 (probability $\leq \frac{2q_E}{2^n}$). The union bound of these disjoint events yields $\frac{2(j+q_E)}{2^n}$.

Applying the triangle inequality, we get:

$$|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| \leq 8\sqrt{\frac{q_P}{2^n}} + \frac{3j}{2^n} + \frac{2q_E}{2^n}.$$

□

Lemma 6 *Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E$, we have:*

$$|\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H_j^*) = 1]| \leq 4\sqrt{\frac{q_P}{2^n}}.$$

Proof We can write the oracle sequences defined by H_j and H_j^* as follows:

$$\begin{aligned} H_j : & \underbrace{(P_1, P_2, R)^j}_{j \text{ classical queries}}, \underbrace{(\mathbf{P}_{1,T_j,k}, P_{2,T_j,k}, E_k[\mathbf{P}_{1,T_j,k}, P_{2,T_j,k}])^{q_E-j}}_{q_E-j \text{ remaining queries}} \\ H_j^* : & \underbrace{(P_1, P_2, R)^j}_{j \text{ classical queries}}, \underbrace{(\mathbf{P}_{1,T_j,k}^{(1)}, P_{2,T_j,k}, E_k[\mathbf{P}_{1,T_j,k}^{(1)}, P_{2,T_j,k}])^{q_E-j}}_{q_E-j \text{ remaining queries}} \end{aligned}$$

Construct a distinguisher $\mathcal{D}$ that participates in the resampling experiment for P_1 . Its behavior is divided into two stages:

Phase 1: $\mathcal{D}$ is given quantum access to random permutations P_1, P_2 and samples a random permutation $R \leftarrow \mathcal{P}_n$ (used to answer classical queries). $\mathcal{D}$ runs the adversary $\mathcal{A}$ and answers its queries: quantum queries are answered with P_1, P_2 ; classical queries are answered with R . $\mathcal{D}$ continues running $\mathcal{A}$ until $\mathcal{A}$ submits the $(j+1)$ -th classical query x_{j+1} (let the list of the first j classical queries be $T_j = ((x_1, y_1), \dots, (x_j, y_j))$).

Phase 2: $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ from the resampling experiment (s_0, s_1 are uniformly random) and quantum access to $P_1^{(b)}$, where b is the random bit of the resampling experiment, and

$$P_1^{(b)}(x) = P_1 \circ \text{swap}_{s_0, s_1}(x)$$

. Set the key component $k = s_0 \oplus x_{j+1}$. Answer $\mathcal{A}$'s $(j+1)$ -th classical query:

$$y_{j+1} = E_k[P_{1,T_j,k}^{(b)}, P_2](x_{j+1}) = P_{1,T_j,k}^{(b)}(x_{j+1} \oplus k) \oplus P_2(x_{j+1} \oplus k) \oplus k.$$

Substituting $k = s_0 \oplus x_{j+1}$, this simplifies to $y_{j+1} = P_{1,T_j,k}^{(b)}(s_0) \oplus P_{2,T_j,k}(s_0) \oplus k$. Continue running $\mathcal{A}$, answering remaining queries: classical queries are answered with $E_k[P_{1,T_j,k}^{(b)}, P_{2,T_j,k}]$; quantum queries are answered with the modified permutations $P_1^{(b)}$ and P_2 , where $P_{1,T_j,k}^{(b)} = S_{T_j,P_1,k} \circ P_1^{(b)}$ (the sequence $S_{T_j,P_1,k}$ ensures consistency of $E_k[P_{1,T_j,k}^{(b)}, P_{2,T_j,k}]$ with classical queries); $\mathcal{D}$ outputs $\mathcal{A}$'s output bit.

When $b = 0$, $P_1^{(0)} = P_1$, and thus $\mathcal{D}$ simulates experiment H_j : $\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0] = \Pr[\mathcal{A}(H_j) = 1]$. When $b = 1$, $P_1^{(1)} = P_1 \circ \text{swap}_{s_0,s_1}$, and thus $\mathcal{D}$ simulates experiment H_j^* : $\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1] = \Pr[\mathcal{A}(H_j^*) = 1]$. Applying the Resampling Lemma (Lemma 2), the number of queries $\mathcal{D}$ makes to P_1 in Phase 1 is q_{P_1} , i.e., the number of quantum queries $\mathcal{A}$ makes to P_1 in Phase 1. Since $\mathcal{A}$'s total quantum queries are q_P , distributed between P_1 and P_2 , we have $q_{P_1} \leq q_P$. Combined with the conclusion of the Resampling Lemma, we get:

$$\begin{aligned} |\Pr[\mathcal{A}(H_j) = 1] - \Pr[\mathcal{A}(H_j^*) = 1]| &= |\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0] - \Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1]| \\ &\leq 4\sqrt{\frac{q_{P_1}}{2^n}} \leq 4\sqrt{\frac{q_P}{2^n}}. \end{aligned}$$

□

Lemma 7 *Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E$, we have:*

$$|\Pr[\mathcal{A}(H_j^*) = 1] - \Pr[\mathcal{A}(H_j^{**}) = 1]| \leq 4\sqrt{\frac{q_P}{2^n}}.$$

Proof We can write the oracle sequences defined by H_j^* and H_j^{**} as follows:

$$\begin{aligned} H_j^* : & \underbrace{(P_1, P_2, R)^j}_{j \text{ classical queries}}, \underbrace{(P_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}, E_k[P_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}])^{q_E-j}}_{q_E-j \text{ remaining queries}} \\ H_j^{**} : & \underbrace{(P_1, P_2, R)^j}_{j \text{ classical queries}}, \underbrace{(P_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}, E_k[P_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}])^{q_E-j}}_{q_E-j \text{ remaining queries}} \end{aligned}$$

Construct a distinguisher $\mathcal{D}$ that participates in the resampling experiment for P_2 . Its behavior is divided into two stages:

Phase 1: $\mathcal{D}$ is given quantum access to random permutations P_1, P_2 and samples a random permutation $R \leftarrow \mathcal{P}_n$ (used to answer classical queries). $\mathcal{D}$ runs the adversary $\mathcal{A}$ and answers its queries: quantum queries are answered with P_1, P_2 ; classical queries are answered with R . $\mathcal{D}$ continues running $\mathcal{A}$ until $\mathcal{A}$ submits the $(j+1)$ -th classical query x_{j+1} . Let $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$ be the list of the first j classical queries.

Phase 2: $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ from the resampling experiment (s_0, s_1 are uniformly random) and quantum access to $P_2^{(b)}$ (where b is the random bit of the resampling experiment). Set the key component $k = s_0 \oplus x_{j+1}$. $\mathcal{D}$ answers $\mathcal{A}$'s $(j+1)$ -th classical query:

$$y_{j+1} = E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(b)}] = P_{1,T_j,k}^{(1)}(x_{j+1} \oplus k) \oplus P_{2,T_j,k}^{(b)}(x_{j+1} \oplus k) \oplus k.$$

Substituting $k = s_0 \oplus x_{j+1}$, this simplifies to $y_{j+1} = P_{1,T_j,k}^{(1)}(s_0) \oplus P_{2,T_j,k}^{(b)}(s_0)$. $\mathcal{D}$ continues running $\mathcal{A}$: it answers remaining classical queries with $E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(b)}]$ and quantum queries with the modified permutations $P_1^{(1)}$ and $P_2^{(b)}$. $\mathcal{D}$ outputs $\mathcal{A}$'s output bit.

When $b = 0$: $P_2^{(0)} = P_2$. Then $\mathcal{D}$ simulates experiment H_j^* : $\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0] = \Pr[\mathcal{A}(H_j^*) = 1]$. When $b = 1$: $P_2^{(1)} = P_2 \circ \text{swap}_{s_0, s_1}$. Then $\mathcal{D}$ simulates experiment H_j^{**} : $\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1] = \Pr[\mathcal{A}(H_j^{**}) = 1]$. By the Resampling Lemma, the number of queries $\mathcal{D}$ makes to P_2 in Phase 1 is q_{P_2} , i.e., the number of quantum queries $\mathcal{A}$ makes to P_2 in Phase 1. Since $\mathcal{A}$'s total quantum queries are q_P , distributed between P_1 and P_2 , we have $q_{P_2} \leq q_P$. Therefore:

$$\begin{aligned} |\Pr[\mathcal{A}(H_j^*) = 1] - \Pr[\mathcal{A}(H_j^{**}) = 1]| &= |\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0] - \Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1]| \\ &\leq 4\sqrt{\frac{q_{P_2}}{2^n}} \leq 4\sqrt{\frac{q_P}{2^n}}. \end{aligned}$$

□

Lemma 8 *Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E$, we have:*

$$|\Pr[\mathcal{A}(H_j^{**}) = 1] - \Pr[\mathcal{A}(H_j^{***}) = 1]| \leq \frac{j}{2^n}.$$

Proof

$$\begin{aligned} H_j^{**} &: \underbrace{(P_1, P_2, R)^j}_{j \text{ queries}}, \underbrace{\mathbf{E}_k[\mathbf{P}_{1, T_j, k}^{(1)}, \mathbf{P}_{2, T_j, k}^{(1)}]}_{(j+1)\text{-th query}}, \underbrace{(P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}, E_k[P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}])^{q_E - j - 1}}_{q_E - j - 1 \text{ remaining queries}} \\ H_j^{***} &: \underbrace{(P_1, P_2, R)^j}_{j \text{ queries}}, \underbrace{\mathbf{R}}_{(j+1)\text{-th query}}, \underbrace{(P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}, E_k[P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}])^{q_E - j - 1}}_{q_E - j - 1 \text{ remaining queries}} \end{aligned}$$

Construct a distinguisher $\mathcal{D}$ whose behavior is divided into two stages:

Phase 1: $\mathcal{D}$ is given quantum access to random permutations P_1, P_2 and samples a random permutation $R \leftarrow \mathcal{P}_n$ (used to answer classical queries). $\mathcal{D}$ runs the adversary $\mathcal{A}$ and answers its queries: quantum queries are answered with P_1, P_2 ; classical queries are answered with R . $\mathcal{D}$ continues running $\mathcal{A}$ until $\mathcal{A}$ submits the $(j+1)$ -th classical query x_{j+1} . Let $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$ be the list of the first j classical queries.

Phase 2: $\mathcal{D}$ receives s_0, s_1 from the resampling experiment (uniformly random). And quantum access to $P_{1, T_j, k}^{(1)}$ and $P_{2, T_j, k}^{(1)}$ (b is the random bit of the resampling experiment). Set the key component $k = s_0 \oplus x_{j+1}$. $\mathcal{D}$ continues running $\mathcal{A}$: it answers the remaining queries: for the $(j+1)$ -th classical query x_{j+1} , use $E_k[P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}]$ to answer; H_j^{***} uses the random function R to answer. The other queries are identical in H_j^{**} and H_j^{***} . Classical queries are answered with $E_k[P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}]$, and quantum queries are answered with the modified permutations $P_{1, T_j, k}^{(1)}$ and $P_{2, T_j, k}^{(1)}$.

We need to prove that in H_j^{**} , the response y_{j+1} for the $(j+1)$ -th query is sufficiently uniformly random from the adversary's perspective, and is indistinguishable from the response using the random function in H_j^{***} .

In H_j^{**} ,

$$\begin{aligned} y_{j+1} &= E_k[P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}](x_{j+1}) = P_{1, T_j, k}^{(1)}(x_{j+1} \oplus k) \oplus P_{2, T_j, k}^{(1)}(x_{j+1} \oplus k) \oplus k, \\ P_{1, T_j, k}^{(1)}(x_{j+1} \oplus k) &= P_{1, T_j, k}^{(1)}(s_0) = P_{1, T_j, k}(s_1), P_{2, T_j, k}^{(1)}(x_{j+1} \oplus k) = P_{2, T_j, k}^{(1)}(s_0) = P_{2, T_j, k}(s_1) \end{aligned}$$

Let $S = \{a_i = x_i \oplus k, 1 \leq i \leq j\}$,

$$P_{1,T_j,k}(s_1) = \begin{cases} u_i, & \text{if } s_1 \in S, s_1 = a_i \\ P_1(s_1), & \text{if } s_1 \notin S \end{cases},$$

$$P_{2,T_j,k}(s_1) = \begin{cases} y_i \oplus u_i \oplus k, & \text{if } s_1 \in S, s_1 = a_i \\ P_2(s_1), & \text{if } s_1 \notin S \end{cases},$$

where u_i is the random value set when reprogramming P_1 .

In H_j^{***} , $y_{j+1} = R(x_{j+1})$ is a random value. Due to the randomness of s_1 , we have $\Pr[s_1 \in S] \leq \frac{j}{2^n}$.

Now analyze the randomness of y_{j+1} .

When $s_1 \notin S$: $P_{1,T_j,k}(s_1) = P_1(s_1)$ (not reprogrammed); $P_{2,T_j,k}(s_1) = P_2(s_1)$ (not reprogrammed). Since P_1, P_2 are random permutations, $P_1(s_1)$ and $P_2(s_1)$ are uniformly random, and combined with the uniformly random k , y_{j+1} is uniformly random.

When $s_1 \in S$: Reprogramming sets $P_{1,T_j,k}$ to some random value u_i and $P_{2,T_j,k}$ to $y_i \oplus u_i \oplus k$. In this case, y_{j+1} in H_j^{**} is $P_{1,T_j,k}(s_1) \oplus P_{2,T_j,k}(s_1) \oplus k = u_i \oplus (y_i \oplus u_i \oplus k) \oplus k = y_i$, which is identical to a result obtained from a previous query, allowing the adversary to significantly distinguish H_j^{**} from H_j^{***} .

Therefore, we can give a specific upper bound:

$$|\Pr[\mathcal{A}(H_j^{**}) = 1] - \Pr[\mathcal{A}(H_j^{***}) = 1]| \leq \Pr[s_1 \in S] \leq \frac{j}{2^n}.$$

□

Lemma 9 Let $\mathcal{A}$ be an adversary making at most q_E classical queries and q_P quantum queries. Then for $j = 0, \dots, q_E - 1$, we have:

$$|\Pr[\mathcal{A}(H_j^{***}) = 1] - \Pr[\mathcal{A}(H'_j) = 1]| \leq \frac{2(j + q_E)}{2^n}.$$

Proof We can write the oracle sequences defined by H_j^{***} and H'_j as follows:

$$H_j^{***} : \underbrace{(P_1, P_2, R)^{j+1}}_{j+1 \text{ classical queries}}, \underbrace{(\mathbf{P}_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}^{(1)}, E_k[\mathbf{P}_{1,T_j,k}^{(1)}, \mathbf{P}_{2,T_j,k}^{(1)}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining queries}}$$

$$H'_j : \underbrace{(P_1, P_2, R)^{j+1}}_{j+1 \text{ classical queries}}, \underbrace{(\mathbf{P}_{1,T_{j+1},k}, \mathbf{P}_{2,T_{j+1},k}, E_k[\mathbf{P}_{1,T_{j+1},k}, \mathbf{P}_{2,T_{j+1},k}])^{q_E-j-1}}_{q_E-j-1 \text{ remaining queries}}$$

The key to distinguishing hybrid experiments H_j^{***} and H'_j is to prove that strict permutation equality holds when certain bad events do not occur. Specifically, we need to show $P_{1,T_j,k}^{(1)} = P_{1,T_j,k}$ and $P_{2,T_j,k}^{(1)} = P_{2,T_j,k}$, where $P_{1,T_j,k}^{(1)}$ and $P_{2,T_j,k}^{(1)}$ are permutations defined based on the first j queries and the resampling points s_0, s_1 , while $P_{1,T_j,k}$ and $P_{2,T_j,k}$ are defined based on the first $j+1$ queries.

Let $S = \{a_i = x_i \oplus k \mid 1 \leq i \leq j\}$.

Define the following bad events: * Bad₀: $s_1 \in S$. Suppose $s_1 = a_i = x_i \oplus k, 1 \leq i \leq j$, where s_1 is the random point used in the reprogramming of P_1 and P_2 . In this case, $P_{1,T_j,k}(s_1) = u_i$, $P_{2,T_j,k}(s_1) = u_i \oplus y_i \oplus k$, and

$$E_k[P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}](x_{j+1}) = P_{1,T_j,k}(s_1) \oplus P_{2,T_j,k}(s_1) \oplus k = y_i,$$

which conflicts with the adversary's previous query result (x_i, y_i) . * Bad₁: $s_0 \in S$ (since s_0 is randomly sampled, i.e., $x_{j+1} \oplus k \in S$, suppose $x_{j+1} = x_i$). In this case,

$$\begin{aligned} P_{1,T_j,k}(s_0) &= P_{1,T_j,k}(x_{j+1} \oplus k) = P_{1,T_j,k}(x_i \oplus k) = u_i, \\ P_{2,T_j,k}(s_0) &= P_{2,T_j,k}(x_{j+1} \oplus k) = P_{2,T_j,k}(x_i \oplus k) = u_i \oplus y_i \oplus k, \\ E_k[P_{1,T_j+1,k}, P_{2,T_j+1,k}](x_{j+1}) &= y_i, \end{aligned}$$

which also conflicts with the previous query result (x_i, y_i) . * Bad₂: The adversary queries a point x in the second phase such that $x \oplus k \in \{s_0, s_1\}$. Since s_0 and s_1 are uniformly random and k is uniformly random, for any adversary query x , we have $\Pr[x \oplus k = s_0] \leq \frac{1}{2^n}$ and $\Pr[x \oplus k = s_1] \leq \frac{1}{2^n}$. Therefore, for q_E classical queries, $\Pr[\text{Bad}_2] \leq \frac{2q_E}{2^n}$.

The overall bad event is $\text{Bad} = \text{Bad}_0 \cup \text{Bad}_1 \cup \text{Bad}_2$, where:

$$\begin{aligned} \Pr[\text{Bad}_0] &\leq \frac{j}{2^n} && (\text{from } s_1 \in S), \\ \Pr[\text{Bad}_1] &\leq \frac{j}{2^n} && (\text{from } s_0 \in S), \\ \Pr[\text{Bad}_2] &\leq \frac{2q_E}{2^n} && (\text{from query conflicts}). \end{aligned}$$

Thus,

$$|\Pr[\mathcal{A}(\text{H}_j^{***}) = 1] - \Pr[\mathcal{A}(\text{H}'_j) = 1]| \leq \Pr[\text{Bad}] \leq \frac{2j}{2^n} + \frac{2q_E}{2^n} = \frac{2(j + q_E)}{2^n}.$$

□

4 Conclusion and Future Work

The development of quantum computing poses a serious threat to symmetric cryptography based on fixed key lengths. To address this, this paper studies the security of the SoEM21 pseudorandom function construction under the quantum Q1 model, which captures the realistic scenario where the attacker only has classical access to the encryption/decryption oracles. By comprehensively applying the Reprogramming and Resampling Lemmas, we have for the first time established a rigorous Q1 security proof for the SoEM21 construction, demonstrating that it has a tight security lower bound of $n/3$ bits. This result matches the known optimal attack complexity, thereby confirming that the SoEM21 construction can still provide a reliable security foundation in the quantum era.

Worthwhile future work includes: generalizing this proof framework to generalized SoEM constructions with a number of branches $t > 2$; investigating whether there exists a provable security bound for SoEM under the stronger Q2 model; or applying the analytical methods developed in this work to the security verification of other permutation-based "summation-type" cryptographic components.

Abbreviations

SoEM, Sum of Even-Mansour; PRF, Pseudorandom Function; QROM, Quantum Random Oracle Model; Q1, Quantum-query Model 1; Q2, Quantum-query Model 2.

Author contributions

YanJin Tan and JunTao Gao conceived the study and developed the theoretical framework. YanJin Tan performed the core security proofs and wrote the main manuscript text. XueLian Li assisted in the formal analysis and verified the mathematical derivations. JunTao Gao supervised the project and critically revised the manuscript for important intellectual content. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

The Author confirms: that the work described has not been published before; that it is not under consideration for publication elsewhere; that its publication has been approved by all co-authors.

Competing interests

The authors declare no competing interests.

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